

Working capital (mis)management – impact of executive age

Hari P. Adhikari^a , Marcin W. Krolikowski^b, James Malm^c,
Nilesh B. Sah^d

^a*Department of Accounting, Economics, Finance and Information Sciences, David B. O'Maley College of Business, Embry-Riddle Aeronautical University, Daytona Beach, FL, USA*

^b*Finance Department, School of Business, Providence College, Providence, RI, USA*

^c*Department of Finance, College of Charleston, Charleston, SC, USA*

^d*Department of Finance and Economics, Gary W. Rollins College of Business, University of Tennessee at Chattanooga, Chattanooga, TN, USA*

Abstract

Several recent studies have used the upper echelons theory to explain the impact of personal traits of top executives on various corporate policies. In this, first of its kind, study we find that older executives invest more in working capital; take longer to convert inventories to cash; and pay their suppliers sooner. These findings are consistent with the argument that risk aversion increases with executive age. Our findings indicate that executive age has significant bearing on working capital management policies. This study also initiates new avenues in research relating behavioral aspect of executives with short-term financial management.

Key words: Working capital management; Cash conversion cycle; Executive age; Trade credit

JEL classification: G30, G40

doi: 10.1111/acf.12591

1. Introduction

Lately, academic studies have presented evidence supporting the idea that corporate policies can be driven by individual personality traits of top executives. Studies that resonate with this idea include Gibbons and Murphy (1992), Prendergast and Stole (1996); Barker and Mueller (2002), Bertrand and Schoar (2003), Davidson *et al.* (2007), Huang *et al.* (2012), Yim (2013), Serfling

Please address correspondence to Hari P. Adhikari via email: adhikarh@erau.edu

(2014) and Adhikari *et al.* (2015). We pursue this direction of research and investigate the unexplored association between the behavioural aspect of the top executives and the firm's short-term financial management. Specifically, combining the arguments presented by the 'Upper Echelons Research' (Hambrick and Mason, 1984; Hambrick, 2007; Chatterjee and Hambrick, 2007) with the evidence presented in several other studies (Vroom and Pahl, 1971; Prendergast and Stole, 1996; Barker and Mueller, 2002; Sundaram and Yermack, 2007; Yim, 2013; Serfling, 2014; Olsen *et al.*, 2016), we view an executive's age as a proxy for risk-taking behaviour and study the relationship between the age of the firm's top executives (Chief Executive Officer (CEO) and Chief Financial Officer (CFO)) and the firm's working capital management.

Conventional wisdom suggests that managers need to expend significant resources towards working capital management in order to ensure efficient and smooth functioning of the firm. Aktas *et al.* (2015) report that a Compustat sample of US firms in 2011 had invested about 24 percent of their total sales and about 18 percent of their book values of assets in working capital management. Similarly, Kieschnick *et al.* (2013) indicate that more than 27 percent of the total assets of US firms is invested in net operating working capital. Additionally, as working capital management directly influences the liquidity of the firm, we cannot underestimate its significance. Furthermore, we periodically find important news articles related to working capital management in the popular press as well. For example, a *Forbes Magazine* article¹ compares the cash conversion cycle of Apple and Samsung and commends Apple for its shorter cash conversion cycle. Similarly, the *Financial Times* reports² that the top 1,000 European companies have expended significant resources in order to improve their use of working capital. This evidence clearly suggests the key role of working capital management in corporate financial management.

Several existing studies provide evidence that cash invested in working capital has a negative relationship with firm profitability (Kim and Chung, 1990; Shin and Soenen, 1998; Wang, 2002; Deloof, 2003; García-Teruel and Martínez-Solano, 2007; Gill *et al.*, 2010; Huynh and Su, 2010; Hayajneh and Yassine, 2011). Along similar lines, Kieschnick *et al.* (2013) find that, on average, firms holding a dollar in cash is worth more than an additional dollar invested in net operating working capital. Similarly, García-Teruel and Martínez-Solano (2007) confirm that firm value increases by minimising the cash conversion cycle.

As investment in working capital is associated with the liquidity of a firm (Ding *et al.*, 2013); lesser investment in working capital may improve the

¹See 'Why is Apple's cash conversion cycle significantly shorter than Samsung's?', *Forbes* (7 June 2016).

²See 'Top 1,000 European companies improve use of working capital', *Financial Times* (24 July 2016).

liquidity position of the firm (faster conversion of inventory and account receivables into cash) but it may also affect sales (less trade credit offered to customers). Firms can also decrease their investment in working capital by delaying payments on their short-term liabilities. Correspondingly, investment in working capital will increase if a firm pays its account payables and other short-term debts more quickly. The balance in liquidity, that is the right amount of investment in working capital, is critical as excess liquidity will reduce the magnitude of a firm's profit (Bhattacharya, 2001), while inadequate liquidity may lead a firm towards bankruptcy (Dunn and Cheatham, 1993), even if the firm is solvent.

Chava and Purnanandam (2010) provide evidence that CEOs and CFOs have substantial influence on the policies of corporate firms. The Securities and Exchange Commission (SEC) also recognises the top role of CEOs and CFOs and requires them to personally verify their companies' financial statements (Huang *et al.*, 2012). For these reasons, we study the risk-taking attitude of CEOs and CFOs (among other executives) in this study.

So, how does the age of the top executives influence their firm's short-term financial management? This is an important question which we explore in this study. While the relationship of executives' age with some of the other corporate decisions such as investments, capital structure or payout policy has already been explored, we know little about the association between executives' age and working capital management.

Several studies provide evidence that executive age impacts their risk-taking behaviour. Vroom and Pahl (1971) report that older managers are more conservative relative to younger ones whereas Prendergast and Stole (1996) and Barker and Mueller (2002) find that younger CEOs often take greater risks when investing compared to older CEOs. Sundaram and Yermack (2007) also indicate that executives become more conservative with an increase in age. Yim (2013) points towards the positive relationship between executive age and risk aversion by looking at acquisitions. Similarly, Serfling (2014) finds that older CEOs pursue less risky investment policies and reduce firm risk through such policies.

We focus primarily on the cash conversion cycle as a measure of working capital management. The cash conversion cycle is a popular measure of working capital management and is defined as the time lag between the collection of revenue from the sales of finished products or services and the payment of expenditures for the purchase of raw materials. It is clear that a longer cash conversion cycle increases the cash investment of a firm. That is, a longer cash conversion cycle increases the firm's investment in working capital.

Using a sample of US firms and observations from 1993 to 2014, we provide evidence that older executives have longer cash conversion cycles. That is, they invest heavily in working capital. Further, we establish that older executives maintain larger inventories and pay their short-term obligations relatively faster than their younger counterparts. Our results are robust to various

alternative model specifications and we also perform robustness tests with an alternative measure of working capital management. Thus, our results lend support to the idea that older executives pursue risk-reducing corporate policies while managing the working capital of their firms which may destroy value for their shareholders.

Our study contributes to the working capital management literature by providing empirical evidence that short-term financial management policies are impacted by the age of the firm's CEO and CFO. Unlike other studies that relate working capital management with only firm characteristics, we are the first to identify its relationship with personal characteristics of top executives. One possible implication of our results would be that older executives (who invest more heavily in working capital) could make their firms less profitable. Also, firms which use trade credit extensively like young, high-growth, low-tangibility and high-volatility firms (Abdulla *et al.*, 2015) should be cautious about the age group of their top management. Our findings can also help initiate new avenues in research relating short-term financial decision making to top executives' characteristics. An interesting topic to explore would be to look at how overconfident executives execute their short-term financial management policies.

The remainder of the paper is organised as follows. Section 2 reviews the literature and develops our hypothesis. Section 3 describes the sample. Section 4 details methodology and reports results and several tests of robustness. Lastly, Section 5 concludes.

2. Literature review and hypothesis development

Alongside long-term financial decisions such as debt and payout policies, where most of the existing research is focused, short-term financing decisions are also very important to the firm. There are a number of studies that analyse the relationship between working capital management and the availability of financing. Jensen and Meckling (1976) and Myers and Majluf (1984) argue that, due to market imperfection, outside capital is costlier than internally generated funds. Stiglitz and Weiss (1981) posited that this might lead to credit rationing. Further, Petersen and Rajan (1997) provided evidence that firms with better access to credit from financial institutions provide trade credit to firms that do not have sufficient access to the same funds. Similarly, the findings of Fazzari and Petersen (1993) suggest that investment in working capital is very sensitive to financing constraints. So, the literature clearly suggests that firms need to pay close attention to their short-term financial management.

Although short-term financial management is critical to a firm, existing research indicates that cash invested in working capital has a negative impact on firm profitability. The implications of the theoretical model by Schiff and Lieber (1974) suggest that a firm's credit and inventory policies (indicated by working capital management) affect the firm's profit. Kim and Chung (1990)

applied an NPV criterion to jointly evaluate inventory and account receivables. Their general conclusion is that investment in working capital management has a negative influence on firm performance. Shin and Soenen (1998) find a negative correlation between a firm's net-trade cycle and profitability and a positive relationship between shorter net trade cycles and risk-adjusted stock returns. Along the same lines, while examining Japanese and Taiwanese firms, Wang (2002) found that firms with higher values invest relatively less significantly in working capital, while firms with lower values invest relatively more in working capital. Deloof (2003) studied Belgian firms from 1991 to 1996 and provided evidence that proper working capital management creates value for a firm. Similarly, the findings of Deloof and Jegers (1996), Padachi (2006), García-Teruel and Martínez-Solano (2007), Gill *et al.* (2010), Huynh and Su (2010), Nobanee *et al.* (2011) and Hayajneh and Yassine (2011), by and large, indicate the negative relationship between investment in working capital and firm value. Based on the valuation model of Faulkender and Wang (2006), Kieschnick *et al.* (2013) conclude that, normally, a dollar cash holding adds higher value to the firm than an extra dollar invested in working capital. They also find evidence that, on average, excess stock returns decreased for those firms that increased net operating working capital. They further demonstrated that excess returns suffered more for firms with limited access to external financing. These results do not mean that any initial investment in working capital will have a detrimental effect on the firm; rather, these results suggest that any additional investment (above a certain level) generally affects firms negatively. In this line, Baños-Caballero *et al.* (2014) study the relationship between the investment in working capital and firm performance more comprehensively and conclude an inverted U-shaped relationship between them. As discussed above, this finding suggests that increase in investment when the investment in working capital management is at a low level helps firms but above a certain level, any further investment in working capital will hurt the firm.

Thus, it can be safely said that working capital management decisions are extremely important for the smooth and efficient functioning of a firm. Managers need to decide on the optimal amount of investment in working capital as investing too much may lead to reduction in profits and firm value, while investing too little may cause liquidity problems.

We focus primarily on the cash conversion cycle as a measure of working capital management. The cash conversion cycle is a popular measure of working capital management (e.g., Deloof, 2003; Padachi, 2006; García-Teruel and Martínez-Solano, 2007; Baños-Caballero *et al.*, 2010) and is defined as the time lag between the collection of revenue from the sales of finished products or services and the payment of expenditures for the purchase of raw materials. It is clear that a longer cash conversion cycle increases the cash investment of a firm. That is, a longer cash conversion cycle increases the firm's investment in working capital. Cash Conversion Cycle comprises three components: Days

Inventory Outstanding, Days Receivables Outstanding and Days Payables Outstanding. A conservative working capital policy would not only try to increase Days Inventory Outstanding and Days Receivables Outstanding (to boost sales) but also try to decrease Days Payables Outstanding at the same time by quickly paying the suppliers. The cumulative impact of these actions would increase the length of Cash Conversion Cycle. Alternatively, an aggressive (risky) working capital policy would not only try to decrease Days Inventory Outstanding and Days Receivables Outstanding but also try to increase Days Payables Outstanding at the same time by delaying payments to suppliers. The cumulative impact of these actions would decrease the length of Cash Conversion Cycle.

‘Upper Echelons Research’ (Hambrick, 2007; Chatterjee and Hambrick, 2007) suggests that corporate policies are impacted by personal traits of top executives. There is an extant literature on the relationship between age of the top executives and their risk-taking behaviour.

Although the proponents of career concern models (Scharfstein and Stein, 1990; Hirshleifer and Thakor, 1992; Zwiebel, 1995; Holmström, 1999; Hunter and Kemp, 2004) argue that younger executives are more risk averse, fearing that their underperformance will harm their reputation severely, there are theories such as quiet life theory, certainty effect theory and some strong evidence in neuroscience research which corroborate that risk-taking behaviour declines with age. Extant empirical finance research also provides a plethora of evidence suggesting that older CEOs are more risk averse compared to their younger counterparts. Quiet life theory suggests that CEOs prefer a quiet life and the preference for this increases with age (Bertrand and Mullainathan, 2003). Harman (1991) and Roberts and Rosenberg (2006) find evidence that the energy level of people decreases with age, corroborating quiet life theory. Invented by Kahneman and Tversky (1979), certainty effect theory posits that if there are options, individuals’ exhibit risk aversion in gains and risk seeking in losses. Studying younger versus older adults in the context of certainty effect theory, Mather *et al.* (2012) find that older adults demonstrate greater preference for certain gains than risky gains, suggesting the decrease in risk-taking behaviour with age. Similarly, neuroscience research also suggests that, because of various physiological and psychological changes, risk-taking behaviour decreases with age (Grady *et al.*, 2000). With age, various personal traits including learning ability, ambition and energy change and affect the risk-taking behaviour of an individual. For example, Kuhnen and Knutson (2011) provide evidence that ‘visual cues inducing anxiety (meant to increase activation in the anterior insula of the brain) make subjects less likely to invest in risky assets.’

The following empirical evidence further supports that with increase in age executives of corporate firms become more risk averse. Vroom and Pahl (1971) report that older managers are more conservative relative to younger ones, whereas Prendergast and Stole (1996) and Barker and Mueller (2002) find that

younger CEOs often take greater risks while investing compared to older CEOs. Similarly, studying 500 top-level business executives, MacCrimmon and Wehrung (1986) provide evidence that risk aversion increased in executives with maturity. Li *et al.* (2017) find that in order to support their qualifications to manage, younger CEOs follow more active investment strategies. Sundaram and Yermack (2007) also suggest that executives become more conservative with an increase in age. Yim (2013) establishes the positive relationship between executive age and risk aversion by studying acquisitions. Similarly, Serfling (2014) finds that older CEOs pursue less risky investment policies and reduce firm risk through such policies. In fact, Serfling (2014) studies major corporate policies such as stock return volatility, investment in R&D, firm diversification and operating leverage, and provides very convincing evidence that CEOs exhibit decreasing risk-taking behaviour as they age. Many other studies providing evidence on the negative relationship between age and risk taking include Morin and Suarez (1983), Albert and Duffy (2012), Arora and Kumari (2015) and Dohmen *et al.*, 2017).

In this study, we attempt to extend the literature on short-term financial management policies by focusing on the determinants (related to personal characteristics of top executives) of working capital management decisions. Although other unit heads lead their units, the CEOs and CFOs significantly impact the corporate policies (Chava and Purnanandam, 2010). Hence, we focus on CEOs and CFOs in this study.

So, the increase in risk aversion with age would suggest that investment in working capital management increases with executive age. Specifically, we would be able to observe that:

- Cash Conversion Cycle (CCC) increases with executive age.
- Days Inventory Outstanding (DIO) increases with executive age.
- Days Receivables Outstanding (DRO) increases with executive age.
- Days Payables Outstanding (DPO) decreases with executive age.

3. Sample description

We construct financial variables from the Compustat annual database. CEO and CFO related variables are from the ExecuComp database of Compustat. Both samples consist of US firms from 1993 to 2014. We exclude financial and utility firms from our analysis.

The full sample of data consists of annual observations of executive age and executive tenure from 1993 to 2014. In our analyses, the dependent variables are *Cash Conversion Cycle*, *Days Inventory Outstanding*, *Days Receivables Outstanding* and *Days Payables Outstanding*. Our main variable of interest is *Executive Age*. Based on prior literature (Hill *et al.*, 2010), we have included the following control variables in our analysis, namely: *Sales Growth*, *Sales*

Volatility, Gross Profit Margin, Operating Cash Flows, Firm Size, Market-to-Book Ratio, Leverage, Tangibility and Distress dummy. To extend our analysis, we also introduce other variables like *Firm Age, Executive Gender and Executive Tenure* in our model specifications. Furthermore, we have performed robustness tests using *Networking Capital Ratio* as an alternative measure of working capital management. We provide the details of all of the variables in the Appendix.

4. Methodology and results

4.1. Univariate results

Table 1A reports the descriptive statistics (mean and median) for the variables used in this study. For the full sample, we observe that the mean (median) days for *Cash Conversion Cycle (CCC)* is 13.56 (27.33) days. *Days Inventories Outstanding, Days Receivables Outstanding, and Days Payables Outstanding* have means (medians) of 68.88 (55.33), 53.85 (51.84) and 110.77 (76.87) days, respectively. In addition, the mean (median) age of the executives is 54.13 (54) years. We have also reported the standard deviation (SD), Quartile 1 and Quartile 3 statistics in this table. We have winsorised all continuous variables at the 1 percent and 99 percent levels.

In Panel A of Table 1B, we split the sample by executive age and observe the mean and median of working capital management related variables. From univariate observations, we find that for executives younger than 50 years, the mean (median) days of *Cash Conversion Cycle* is -1.40 (17.66). However, for those executives between the ages of 50 and 60, the mean (median) duration of *Cash Conversion Cycle* is 14.62 (28.44) days. Further, for executives older than 60 years, the mean (median) days for *Cash Conversion Cycle* is 26.36 (33.94) days. The difference between the groups for *Cash Conversion Cycle* is statistically significant and this trend clearly indicates that older executives invest more in working capital when compared to their younger peers. We have done a similar univariate analysis (unreported ANOVA tests) for *Days Inventories Outstanding, Days Receivables Outstanding and Days Payables Outstanding*. We find that *Days Inventories Outstanding* increases with executive age and *Days Payables Outstanding* decreases with executive age, implying that older executives take longer to convert inventories into cash, while they disburse short-term obligations fairly quickly. These statistics are consistent with the view that the older executives are more risk averse than younger executives. Furthermore, increase in *Days Inventories Outstanding* with the age of the executive and decrease in *Days Payables Outstanding* with the age of the executive both contribute to an increase in the cash conversion cycle. *Days Receivables Outstanding* does not display a clear trend with age.

In Panels B and C of Table 1B, we repeat the univariate analysis performed in Panel A separately for CEOs and CFOs respectively. The results in both

Table 1A
Summary statistics

Variable	Mean	SD	Quartile 1	Median	Quartile 3
<i>CEO Age</i>	54.13	7.66	49	54	59
<i>CEO Tenure</i>	7.63	7.84	3	6	10
<i>CEO Gender</i>	0.04	0.2	0	0	0
<i>CEO Duality</i>	0.72	0.45	0	1	1
<i>Sales Growth</i>	0.11	0.51	−0.03	0.07	0.17
<i>Sales Volatility</i>	685.7	1614.2	52.01	163.6	527.95
<i>Gross Profit Margin</i>	0.4	0.21	0.24	0.37	0.53
<i>Operating Cash Flow</i>	0.13	0.14	0.09	0.13	0.18
<i>Firm Size</i>	7.23	1.66	6.12	7.09	8.27
<i>Market-to-Book</i>	2.01	1.29	1.22	1.61	2.3
<i>Leverage</i>	0.15	0.15	0.02	0.11	0.22
<i>Tangibility</i>	0.27	0.22	0.1	0.21	0.38
<i>KZ Index</i>	−5.97	13.3	−6.11	−1.85	0.08
<i>Distress Dummy</i>	0.48	0.5	0	0	1
<i>Cash conversion</i>	13.56	123.64	−15.82	27.33	68.29
<i>Days Receivables</i>	53.85	34.33	33.03	51.84	69.73
<i>Days Payable</i>	110.8	121.8	50.93	76.87	117.49
<i>Days Inventory</i>	68.88	70.3	12.68	55.33	97.82

This table presents the univariate statistics for the variables used. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of annual observations between 1993 and 2014. We use 25,439 observations to obtain the following summary statistics. Variable definitions are presented in the Appendix.

specifications are qualitatively similar to the combined sample of CEOs and CFOs.

In Table 1C, we present a correlation matrix between executive age and measures of managerial risk-taking behaviour namely: *R&D expenses* and *Acquisition expenses*. We create this correlation matrix to observe whether risk aversion increases with the CEO age. We have defined *R&D expenses* as the total research and development expenditure of the firm in that year scaled by the firm's total assets. Similarly, we have defined *Acquisition expenses* as the total acquisition expenditure of the firm in that year scaled by the firm's total assets. We include these two variables in our analysis based on the prior literature. R&D expenses are risk-increasing investments and so lower R&D expenses indicate managerial risk aversion (Bhagat and Welch, 1995; Kothari *et al.*, 2002; Eberhart *et al.*, 2004; Coles *et al.*, 2006; Cassell *et al.*, 2012; Kini and Williams, 2012). Similarly, Serfling (2014) and Yim (2013) find that acquisitions are risky investments and lower acquisitions indicate managerial risk aversion. The results in Table 1C show that executive age is significantly negatively correlated with both investment in R&D and investment in acquisitions, suggesting older CEOs are more risk averse. The results of this

Table 1B

Summary statistics for all executives and separately for CEOs and CFOs

Variable	Age less than 50 years	Age between 50 and 60 years	Age greater than 60 years
Panel A: All executives (CEOs and CFOs)	(<i>n</i> = 6,643)	(<i>n</i> = 12,031)	(<i>n</i> = 6,765)
<i>Cash Conversion</i>	−1.40	14.62	26.36
<i>Days Receivables</i>	52.88	53.73	55.00
<i>Days Payable</i>	119.12	110.90	102.34
<i>Days Inventory</i>	63.30	70.08	72.20
Panel B: Only CEOs	(<i>n</i> = 5,205)	(<i>n</i> = 9,427)	(<i>n</i> = 5,301)
<i>Cash Conversion</i>	−5.27	15.13	28.18
<i>Days Receivables</i>	54.96	54.40	55.09
<i>Days Payable</i>	123.33	110.93	101.17
<i>Days Inventory</i>	61.33	69.82	72.63
Panel C: Only CFOs	(<i>n</i> = 1,438)	(<i>n</i> = 2,604)	(<i>n</i> = 1,464)
<i>Cash Conversion</i>	3.27	13.31	17.39
<i>Days Receivables</i>	50.38	51.99	54.53
<i>Days Payable</i>	114.04	110.84	108.11
<i>Days Inventory</i>	65.68	70.76	70.07

This table presents the univariate statistics for the measures of working capital based on age groups. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The sample is split in three parts (<50 years, 50–60 years, and >60 years) based on executive age. Panel A consists of observations for both CEOs and CFOs; Panel B consists of observations for CEOs only; and Panel C consists of observations for CFOs only. The sample consists of annual observations between 1993 and 2014. Variable definitions are presented in the Appendix.

correlation matrix signal older executives as risk averse and motivates us to further explore the relationship between executive age and investment in working capital.

4.2. Multivariate results

In this section, we explore the relationship between executive age and working capital management in a multivariate setting that allows us to control for various other factors. In our main multivariate regression models, we use *Cash Conversion Cycle*, *Days Inventories Outstanding*, *Days Receivables Outstanding* and *Days Payables Outstanding* as the dependent variable.

We employ OLS regressions and regression with standard errors clustered at the firm and executive level. After performing OLS regressions, following Petersen (2009), we cluster standard errors at firm and executive levels to mitigate

Table 1C

Correlation matrix between executive age and measures of managerial risk aversion

	Executive age	R&D expenses (scaled by assets)	Acquisition expenses (scaled by assets)
<i>Executive Age</i>	1.0000		
R&D expenses (scaled by assets)	−0.0307***	1.000	
Acquisition expenses (scaled by assets)	−0.0142***	−0.0071	1.000

This table presents the correlation matrix between the executive age and measures of managerial risk aversion. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of annual observations between 1993 and 2014. Variable definitions are presented in the Appendix.

the bias triggered by correlated residuals. By clustering standard errors at firm and executive level, we avoid the possibility of misleading small standard errors, large t -statistics and small p -values as explained by Cameron and Miller (2015). In multivariate settings, we winsorise all continuous variables at the 1 and 99 percent levels to remove any effect of outliers on the results (Billett *et al.*, 2007). Also, we calculate all dollar values in 2002 dollars.

In Models 1 and 2 of Table 2, we use *Cash Conversion Cycle* as a comprehensive measure of working capital management and examine the relationship between *Executive Age* and *Cash Conversion Cycle* for the overall sample period, 1993–2014. Thus, our dependent variable is *Cash Conversion Cycle* and our main variable of interest is *Executive Age*. In Model 1, we perform OLS regression and in Model 2, we perform regressions with standard errors clustered at the firm and executive level.

The regression model is as follows:

$$\begin{aligned} \text{Cash Conversion Cycle} = f(\text{Executive Age, Sales Growth, Sales Volatility,} \\ \text{Gross Profit Margin, Operating Cash Flow,} \\ \text{Firm Size, } M/B, \text{Leverage,} \\ \text{Tangibility, Distress Dummy}) \end{aligned}$$

In both specifications, we find that *Executive Age* is significantly positively related to the cash conversion cycle of the firm. For example, Model 1 suggests that the cash conversion cycle for an executive who is 10 years older would be 13.34 days longer as compared to a younger executive. *Ceteris paribus*, this difference is significant at the 1 percent level. We can similarly interpret the coefficients in Model 2.

Overall, the results suggest that older executives become more risk averse leading to longer cycles in cash conversion in the firms they lead. The results are

Table 2
Multivariate regressions for cash conversion cycle

	CEOs and CFOs		CEOs only		CFOs only	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	OLS Regression	Standard errors are clustered at firm and executive level	OLS regression	Standard errors are clustered at firm and executive level	OLS regression	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	1.3342*** (0.088)	0.8039*** (0.152)	1.3381*** (0.099)	0.8254*** (0.178)	1.2451*** (0.238)	0.9065*** (0.324)
<i>Sales Growth (t-1)</i>	-5.5053*** (1.317)	-5.5613*** (1.303)	-5.0852*** (1.356)	-5.8266*** (1.331)	-7.8595 (5.046)	-2.2075 (5.329)
<i>Sales Volatility</i>	-0.0030*** (0.001)	-0.0011 (0.001)	-0.0026*** (0.001)	-0.0006 (0.001)	-0.0045*** (0.001)	-0.0027* (0.001)
<i>Gross Profit Margin (t-1)</i>	-240.7585*** (6.260)	-220.4791*** (11.875)	-230.4070*** (7.127)	-210.5161*** (14.231)	-274.6377*** (13.100)	-252.2657*** (20.501)
<i>Operating Cash Flow (t-1)</i>	142.9706*** (9.158)	128.9730*** (13.191)	147.6375*** (10.231)	133.4142*** (15.214)	123.8980*** (20.783)	112.0860*** (26.091)
<i>Firm Size (t-1)</i>	-0.7151 (0.616)	-1.3447 (1.089)	-0.9966 (0.700)	-1.3420 (1.291)	0.5028 (1.300)	-1.3552 (1.862)
<i>Market-to-Book (t-1)</i>	-7.0822*** (0.861)	-5.2324*** (1.366)	-7.2989*** (0.951)	-5.8522*** (1.552)	-6.6293*** (2.085)	-3.3693 (2.633)
<i>Leverage (t-1)</i>	2.3511 (5.765)	7.7355 (9.525)	3.0929 (6.392)	5.5141 (10.900)	-1.0358 (12.909)	16.0630 (18.846)
<i>Tangibility (t-1)</i>	-119.5671*** (4.018)	-68.7797*** (8.213)	-119.1667*** (4.471)	-76.0297*** (9.493)	-123.2432*** (9.190)	-43.7125*** (15.388)
<i>Distress Dummy</i>	5.4425***	1.8776	5.3845***	2.3838	6.3720*	1.3843

(continued)

Table 2 (continued)

Cash conversion cycle	CEOs and CFOs		CEOs only		CFOs only	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	Standard errors are clustered at firm and executive level		Standard errors are clustered at firm and executive level		Standard errors are clustered at firm and executive level	
OLS Regression	(1.625)	(2.525)	(1.800)	(2.957)	(3.747)	(4.644)
Constant	70.9109*** (6.049)	112.8729*** (26.224)	68.2610*** (6.947)	119.0767*** (31.805)	83.6434*** (14.059)	60.7009** (30.087)
R ²	0.194	0.334	0.187	0.326	0.221	0.371
Adj R ²	0.1937	0.3337	0.1866	0.3257	0.2196	0.3699
Observations	25,439	25,439	19,933	19,933	5,506	5,506
Industry controls	No	Yes	No	Yes	No	Yes
Year controls	No	Yes	No	Yes	No	Yes

This table presents the multivariate regressions to find the determinants of *Cash Conversion Cycle* using different methodologies. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for all models are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

also consistent with previous findings such as older executives tend to be more conservative (Hambrick and Mason, 1984), more risk averse (MacCrimmon and Wehrung, 1986) and follow lower growth strategies (Child, 1974) when compared to their younger counterparts. The other control variables are, in general, consistent in magnitude and signs with the existing literature.

In Models 3 and 4 of Table 2, we split the sample and study the impact of age for CEOs. In both models, we still find qualitatively similar result to Models 1 and 2. For example, in Model 4 of Table 2, in which we have clustered the standard errors at firm and executive level, the cash conversion cycle for a 10 years older CEO is approximately 8.25 days longer than for a 10 years younger CEO. In Models 5 and 6 of Table 2, we study the impact of age for CFOs and the cash conversion cycle. In these specifications, also we find similar results. For example, in Model 6 of Table 2, in which we have clustered the standard errors at firm and executive level, the cash conversion cycle for a 10 years older CFO is approximately 9.07 days longer than for a 10 years younger CFO.

In Table 3, we examine the relationship between *Executive Age* and the components of the cash conversion cycle measure. Specifically, in Model 1, we study the effect of *Executive Age* on *Days Receivable Outstanding*. Our dependent variable is *Days Receivables Outstanding*, and the main variable of interest is *Executive Age*. Similarly, in Models 2 and 3 the dependent variables are *Days Payables Outstanding* and *Days Inventory Outstanding*. The model specification and control variables remain the same as in Table 2 and we cluster the standard errors at firm and executive level.

In these specifications, we find that *Executive Age* is significantly positively related to *Days Inventory Outstanding*, suggesting that older executives, on average, take longer to convert inventories into cash, whereas *Executive Age* is significantly negatively related to *Days Payables Outstanding*, implying that older executives do not hold payables longer when compared to their younger counterparts. In other words, they contribute negatively on the liability side of the balance sheet by paying early, thereby increasing the cash conversion cycle. The relation between *Executive Age* and *Days Receivable Outstanding* is not statistically significant in the multivariate setting. The relationship between *Executive Age* and *Days Inventory Outstanding* and *Days Payables Outstanding* are consistent with older executives being risk averse.

Specifically, Model 3 of Table 3 suggests that, on average, it takes approximately 3 days longer for a 10 years older executive to convert inventory into cash. This coefficient of *Executive Age* is significant at the 1 percent level. Whereas Model 2 reports that, on average, a 10 years older executive pays the outstanding accounts payable approximately 4.54 days earlier. Thus, on average, older executives contribute to increasing the cash conversion cycle by allowing a longer time to convert inventory into cash and by paying their accounts payable earlier. These findings suggest that older executives do not

Table 3
Multivariate regressions for the components of cash conversion cycle

All executives			
	<i>Days receivables</i>	<i>Days payable</i>	<i>Days inventory</i>
	Model 1	Model 2	Model 3
	Standard errors are clustered at firm and executive level	Standard errors are clustered at firm and executive level	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	0.0709 (0.045)	-0.4546*** (0.129)	0.3039*** (0.084)
<i>Sales Growth (t-1)</i>	-1.3993*** (0.375)	1.5643 (1.279)	-2.4546*** (0.641)
<i>Sales Volatility</i>	-0.0007* (0.000)	0.0002 (0.001)	-0.0007* (0.000)
<i>Gross Profit Margin (t-1)</i>	16.0833*** (2.824)	344.2437*** (11.530)	104.7658*** (4.798)
<i>Operating Cash Flow (t-1)</i>	-22.7496*** (3.301)	-181.5993*** (13.392)	-22.3787*** (6.784)
<i>Firm Size (t-1)</i>	2.1660*** (0.317)	4.4601*** (1.107)	0.1666 (0.517)
<i>Market-to-Book (t-1)</i>	-0.8011** (0.327)	2.4050* (1.291)	-2.0488*** (0.655)
<i>Leverage (t-1)</i>	12.6365*** (3.835)	29.3508*** (7.240)	16.9854*** (5.524)
<i>Tangibility (t-1)</i>	-45.3256*** (2.586)	-33.0089*** (8.355)	-48.3831*** (4.391)
<i>Distress Dummy</i>	4.0831*** (0.824)	7.7283*** (2.380)	5.2164*** (1.544)
Constant	63.6982*** (12.668)	10.9502 (23.397)	62.6223*** (16.797)
R^2	0.327	0.419	0.474
Adj R^2	0.3267	0.4188	0.4738
Observations	25,439	25,439	25,439
Industry controls	Yes	Yes	Yes
Year controls	Yes	Yes	Yes

This table presents the multivariate regressions to find the determinants of the components of *Cash Conversion Cycle*. The dependent variables are *Days Receivables*, *Days Payable* and *Days Inventory* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Models 1, 2 and 3 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

pursue aggressive short-term financial management policies (remain risk averse) and invest more in working capital.

4.3. Robustness tests

In addition to the multivariate regressions examining the effect of CEO age on the cash conversion cycle and its components, we also assess the robustness of these findings by performing statistical analyses considering executive characteristics (executive tenure and gender of executive) and firm characteristics (firm age and firm type) which might affect the cash conversion cycle. In other words, we expect that the results we find in the multivariate analyses above, that *Cash Conversion Cycle* (CCC) increases with age, are not driven by executive tenure or executive gender nor because of firm age or firm type. To further corroborate our findings, we perform robustness tests using an alternative measure of *Cash Conversion Cycle*. Additionally, we examine the difference in cash conversion cycle when both CEO and CFO are young versus when both are old.

One argument is that executives with long tenure might weigh their reputation more and are less likely to take risks. So, for those executives the benefit of investing less in working capital (that is, shortening the cash conversion cycle) may be less than the cost of retaining reputation. These views are consistent with Diamond (1989) in the context of reputation acquisition in debt markets and Ali and Zhang (2015) in the relationship between CEO tenure and earnings management. So, the tenure of the executives might also have an influence on the cash conversion cycle. Some prior studies include executive tenure as a proxy for risk aversion (Coles *et al.*, 2006; Chakraborty *et al.*, 2007; Cain and McKeon, 2016). In order to rule out the possibility that our results are driven by the effect of the tenure of executives, we divide our sample into two groups (based on executive tenure) and conduct our analysis like the analysis in Li *et al.* (2017). In Model 1 of Table 4A, we include the executives with less than or equal to six years of tenure and in Model 2 we include executives with more than six years of tenure. As mentioned above, we expect that *Executive Age* remains positive and significant in both subsamples. Consistent with our expectation, in both the models, we find that *Executive Age* is significantly positively associated (at the 1 percent level) with the cash conversion cycle suggesting that irrespective of tenure, older executives are related to longer cash conversion cycles. As earlier, in both the specifications we have clustered standard errors at firm and executive levels. In Table 4B, we conduct a difference in mean test between executives with tenure less than or equal to 6 years and executives with tenure of more than 6 years and find that the cash conversion cycle is significantly more (at the 1 percent level) for longer tenured executives than for shorter tenured executives and the components of cash conversion cycles are also different for two groups of executives. This

Table 4A

Multivariate regressions for examining the impact of executive tenure

Cash conversion cycle		
All executives		
	Model 1	Model 2
	Tenure ≤ 6 years	Tenure > 6 years
	Standard errors are clustered at firm and executive level	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	0.6153** (0.241)	0.7609*** (0.184)
<i>Sales Growth (t-1)</i>	-5.7780*** (1.850)	-5.3742*** (1.802)
<i>Sales Volatility</i>	-0.0005 (0.001)	-0.0017 (0.001)
<i>Gross Profit Margin (t-1)</i>	-213.3440*** (15.080)	-226.9445*** (15.429)
<i>Operating Cash Flow (t-1)</i>	129.6092*** (16.712)	127.0325*** (18.617)
<i>Firm Size (t-1)</i>	-0.5784 (1.432)	-1.6980 (1.452)
<i>Market-to-Book (t-1)</i>	-7.7747*** (2.010)	-3.5086** (1.746)
<i>Leverage (t-1)</i>	2.7627 (12.712)	9.9931 (12.628)
<i>Tangibility (t-1)</i>	-75.5556*** (10.446)	-61.6684*** (10.522)
<i>Distress Dummy</i>	1.0852 (3.285)	2.9022 (3.401)
Constant	136.3414*** (38.952)	101.6969*** (28.831)
R^2	0.332	0.343
Adj R^2	0.3314	0.3425
Observations	11,012	14,427
Industry controls	Yes	Yes
Year controls	Yes	Yes

This table presents the multivariate regressions to find the impact of executive tenure. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – executives having tenure less than or equal to 6 years and executives having tenure > 6 years. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Models 1 and 2 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

Table 4B

Difference in mean cash conversion cycle test for executive tenure

Mean cash conversion cycle			
	Tenure $\leq$ 6 years ($n = 11,012$)	Tenure $>$ 6 years ($n = 14,427$)	Difference in means
<i>Cash Conversion</i>	11.22	15.08	***
<i>Days Receivables</i>	54.19	53.62	*
<i>Days Payable</i>	112.04	109.95	*
<i>Days Inventory</i>	67.33	69.88	***

This table presents the difference in mean cash conversion cycle between long tenured and short tenured executives. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – executives with tenure less than or equal to 6 years and executives with tenure more than 6 years. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. Variable definitions are presented in the Appendix.

result, together with our regression results in Table 4A, substantiates the effect of executive age and their risk aversion on the cash conversion cycle.

Prior literature has provided evidence that there are differences between male and female executives in regard to risk taking and, hence, in undertaking various corporate policies (Dittmar and Duchin, 2010; Adhikari, 2012; Huang and Kisgen, 2013; Francis *et al.*, 2015; Hunter *et al.*, 2015; Faccio *et al.*, 2016). Faccio *et al.* (2016) provide evidence that female CEOs are more risk averse than male CEOs. Although there are significantly fewer female executives in US corporations, we still investigate the possibility that the results so far may be driven by risk aversion of female executives. So, we divide the sample into firms with male executives and female executives. Results from Model 1 of Table 5A show that *Executive Age* is highly significant only when we consider male executives. In Model 2, in which we consider only female executives, the *Executive Age* variable is not significant (although it is positive). These results help us to separate the age effect from the gender effect. In Table 5B, we also perform regression analysis by pooling both female and male executives together and using female executive as a dummy variable. In this regression, we also include an interaction term between *Executive Age* and *Executive Gender*. We execute the regression by clustering standard errors at firm and executive levels. The results show that the executive age is highly significantly positively related to cash conversion cycle and the gender dummy and the interaction terms are not significant, further establishing our results.

We also perform a difference in mean test for cash conversion cycle and its components between male and female executives in Table 5C. We observe that the cash conversion cycle for male executives is longer than for female

Table 5A

Multivariate regressions for examining the executive gender (split sample)

Cash conversion cycle		
All executives		
	Model 1	Model 2
	Male	Female
	Standard errors are clustered at firm and executive level	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	0.8372*** (0.154)	0.5603 (0.735)
<i>Sales Growth (t-1)</i>	-5.3713*** (1.310)	-11.9894 (11.457)
<i>Sales Volatility</i>	-0.0010 (0.001)	-0.0080*** (0.002)
<i>Gross Profit Margin (t-1)</i>	-220.3147*** (12.043)	-262.8881*** (68.333)
<i>Operating Cash Flow (t-1)</i>	132.6298*** (13.397)	26.8771 (65.071)
<i>Firm Size (t-1)</i>	-1.3891 (1.115)	0.7904 (4.112)
<i>Market-to-Book (t-1)</i>	-4.8990*** (1.366)	-14.1475 (8.709)
<i>Leverage (t-1)</i>	9.5540 (9.672)	-4.1971 (45.950)
<i>Tangibility (t-1)</i>	-69.5925*** (8.401)	-41.6775 (28.739)
<i>Distress Dummy</i>	1.0393 (2.593)	20.4622** (10.168)
Constant	110.2420*** (26.329)	171.1707** (81.558)
R^2	0.334	0.460
Adj R^2	0.3337	0.4541
Observations	24508	931
Industry controls	Yes	Yes
Year controls	Yes	Yes

This table presents the multivariate regressions to find the impact of executive gender. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – male and female executives. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Models 1 and 2 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

Table 5B

Multivariate regressions for examining the impact of executive gender

Cash conversion cycle	
All executives	
	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	0.8323*** (0.1574)
<i>Executive Gender</i>	55.4947 (41.2042)
<i>Executive Age*Executive Gender</i>	−1.0183 (0.8012)
<i>Sales Growth (t−1)</i>	−5.5407*** (1.3602)
<i>Sales Volatility</i>	−0.0011 (0.0015)
<i>Gross Profit Margin (t−1)</i>	−220.2903*** (16.5507)
<i>Operating Cash Flow (t−1)</i>	128.9296*** (16.2992)
<i>Firm Size (t−1)</i>	−1.3482 (1.5109)
<i>Market-to-Book (t−1)</i>	−5.2346*** (1.6348)
<i>Leverage (t−1)</i>	7.9712 (12.9575)
<i>Tangibility (t−1)</i>	−68.9197*** (11.4556)
<i>Distress Dummy</i>	1.8883 (3.2480)
Constant	111.2017*** (35.5385)
R^2	0.3343
Adj R^2	0.3340
Observations	25,439
Industry controls	Yes
Year controls	Yes

This table presents the multivariate regressions to find the impact of executive gender. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. *Executive Gender* is a dummy variable equal to 1 if the executive is female. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – male and female executives. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Models 1 and 2 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

Table 5C
Difference in mean cash conversion cycle test for executive gender

Mean cash conversion cycle			
	Male (n = 24,508)	Female (n = 931)	Difference in means
<i>Cash Conversion</i>	13.62	12.31	
<i>Days Receivables</i>	54.14	47.23	***
<i>Days Payable</i>	111.15	102.47	**
<i>Days Inventory</i>	68.99	66.45	

This table presents the difference in mean cash conversion cycle and its components between male and female executives. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – male executives and female executives. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. Variable definitions are presented in the Appendix.

executives, even though the difference is not significant. Considering female executives as more risk averse, we would expect the opposite. But, in our sample of 931 female executive observations, 385 are under the age of 50; 491 are between the ages of 50 and 60 and only 55 observations are above the age of 60. Also, the median age for female executives is 53 years for our sample. So, comparing the distribution of age groups in the main sample as mentioned in Table 1B (including both male and female) with the distribution of age groups of females mentioned above, we observe that the female executives are relatively younger than the male executives. So, it is plausible that the age effect is shielding the effect of more risk aversion manifested by female executives.

There are two views on how firm age influences firm activities and outcomes. Studies like Arrow (1962), Jovanovic (1982) and Ericson and Pakes (1995) argue that firms learn over time and become more efficient, resulting in better coordination, cost reduction and quality improvement. On the other hand, articles such as Agarwal and Gort (1996, 2002) argue that older firms are less efficient because of their obsolete knowledge, abilities and skills. One can argue that the cash conversion cycle may vary with firm age as the relationship of mature firms with their buyers and suppliers would be different than the relationship younger firms have with their buyers and suppliers.

So, we divide our sample into young and old firms and conduct our analysis in Models 1 and 2 of Table 6A respectively. Similar to previous results, we find that the older executives are related to the longer cash conversion cycles. However, it is also interesting to note that the impact of executive age is more pronounced for younger firms. This result may be because of lower bargaining power possessed by younger firms as compared to mature firms.

Table 6A
Multivariate regressions for examining the impact of firm age

	Model 1	Model 2
	Young firms	Older firms
	Standard errors are clustered at firm and executive level	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	0.9762*** (0.244)	0.4300*** (0.164)
<i>Sales Growth (t-1)</i>	-7.1027*** (1.816)	-0.4359 (1.433)
<i>Sales Volatility</i>	-0.0051*** (0.002)	-0.0002 (0.001)
<i>Gross Profit Margin (t-1)</i>	-282.5584*** (15.370)	-115.1114*** (14.958)
<i>Operating Cash Flow (t-1)</i>	147.9315*** (15.153)	79.2375*** (24.301)
<i>Firm Size (t-1)</i>	-2.4834 (1.875)	-3.7093*** (1.171)
<i>Market-to-Book (t-1)</i>	-4.6990*** (1.615)	-2.4459 (2.370)
<i>Leverage (t-1)</i>	-23.9035* (13.328)	38.7894*** (12.343)
<i>Tangibility (t-1)</i>	-42.0392*** (10.935)	-90.5240*** (10.880)
<i>Distress Dummy</i>	11.3867*** (3.874)	-6.1436** (2.904)
Constant	146.5105*** (39.940)	108.8848*** (27.492)
R^2	0.395	0.271
Adj R^2	0.3945	0.2704
Observations	12,328	13,111
Industry controls	Yes	Yes
Year controls	Yes	Yes

This table presents the multivariate regressions to find the impact of firm age. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – young firms (firm age less than or equal to 20 years) and older firms (firm age greater than 20 years). ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Models 1 and 2 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

We also perform a difference in mean test for cash conversion cycle and its components between young and old firms in Table 6B. We find positive cash conversion cycles for older firms and negative cash conversion cycles for

Table 6B
Difference in mean cash conversion cycle test for firm age

Mean cash conversion cycle			
	Young firms (<i>n</i> = 12,328)	Old firms (<i>n</i> = 13,111)	Difference in means
<i>Cash Conversion</i>	−4.53	31.18	***
<i>Days Receivables</i>	54.52	53.19	*
<i>Days Payable</i>	125.68	96.25	***
<i>Days Inventory</i>	64.75	72.90	***

This table presents the difference in mean cash conversion cycle and its components between young firms and old firms. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – young firms (firm age less than or equal to 20 years) and older firms (firm age greater than 20 years). ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. Variable definitions are presented in the Appendix.

younger firms, and the difference is also significant. Incorporating this mean test result with regression results in Table 6A, we can infer that the executive age effects on cash conversion cycle are not influenced by the firm age effect.

In Table 7A, we undertake subsample analysis by splitting the entire sample into high-tech firms and labour-intensive firms (based on the high-tech firm classification of Loughran and Ritter, 2004). We consider all firms other than high-tech as labour-intensive firms. Generally, because of the nature of the business, high-tech firms have to invest much less in working capital as compared to other firms. For example, a software company that sells software through its website will have little inventory to maintain and hence will have low investment in working capital. For these reasons, we separately study the effect of executive age on cash conversion cycle for labour-intensive and high-tech firms. However, we find that the cash conversion cycle is significantly longer for older executives in both subsamples. Table 7B demonstrates the difference in mean between the cash conversion cycle and its components, substantiating our analysis.

Finally, we perform some more robustness tests to further explore the relationship between the age of the top executives and working capital management. For that purpose, first we use *Net Working Capital Ratio* as an alternative measure of the investment in short-term financing. We measure *Net Working Capital Ratio* using two definitions. The first definition is the difference between current assets and current liabilities and scale that difference by the total assets of the firm. The second definition is the difference between *Current Assets* and *Accounts Payable* scaled by *Total Assets*. Using these two

Table 7A
Multivariate regressions for examining the impact of firm type

	Labour-intensive firms	Hi-tech firms
	Standard errors are clustered at firm and executive level	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	0.7066*** (0.161)	0.9424** (0.384)
<i>Sales Growth (t-1)</i>	-6.7912*** (1.416)	-0.3959 (3.316)
<i>Sales Volatility</i>	-0.0005 (0.001)	-0.0018 (0.002)
<i>Gross Profit Margin (t-1)</i>	-176.8633*** (13.352)	-295.9443*** (24.034)
<i>Operating Cash Flow (t-1)</i>	126.7735*** (17.129)	85.2165*** (19.618)
<i>Firm Size (t-1)</i>	-0.1267 (1.188)	-5.4383** (2.624)
<i>Market-to-Book (t-1)</i>	-6.2749*** (1.585)	0.9907 (2.667)
<i>Leverage (t-1)</i>	11.8972 (10.012)	-30.1066 (27.659)
<i>Tangibility (t-1)</i>	-87.3842*** (8.752)	23.5321 (27.363)
<i>Distress Dummy</i>	2.3531 (2.854)	1.2479 (5.344)
Constant	106.3483*** (25.661)	221.6575*** (31.212)
R^2	0.311	0.419
Adj R^2	0.3107	0.4179
Observations	20286	5153
Industry controls	Yes	Yes
Year controls	Yes	Yes

In line with Loughran and Ritter (2004), hi-tech stocks are defined as those with SIC codes 3571, 3572, 3575, 3577, 3578 (computer hardware), 3661, 3663, 3669 (communications equipment), 3671, 3672, 3674, 3675, 3677, 3678, 3679 (electronics), 3812 (navigation equipment), 3823, 3825, 3826, 3827, 3829 (measuring and controlling devices), 3841, 3845 (medical instruments), 4812, 4813 (telephone equipment), 4899 (communications services), 7371, 7372, 7373, 7374, 7375, 7378, and 7379 (software) and remaining SIC codes are considered labour intensive.

This table presents the multivariate regressions to find the impact of firm type. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – hi-tech firms and labour-intensive firms. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Models 1 and 2 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

Table 7B
Difference in mean cash conversion cycle test for firm type

Mean cash conversion cycle			
	Labour-intensive firms (<i>n</i> = 20,286)	Hi-tech firms (<i>n</i> = 5,153)	Difference in means
<i>Cash Conversion</i>	19.97	−10.98	***
<i>Days Receivables</i>	50.53	66.55	***
<i>Days Payable</i>	100.41	150.45	***
<i>Days Inventory</i>	68.49	70.37	**

This table presents the difference in mean cash conversion cycle and its components between hi-tech firms and labour-intensive firms. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. The sample is split into two parts – hi-tech firms and labour-intensive firms. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. Variable definitions are presented in the Appendix.

definitions of *Net Working Capital Ratio* as dependent variables in Models 1 and 2 of Table 8A, we repeat the multivariate analysis performed in Table 2. Our variable of interest is *Executive Age*. We find that *Executive Age* is significantly positively related to *Net Working Capital Ratio* in both models. The significance level has remained at the 1 percent level even with these measures of working capital management. As such, our results are robust to the very basic measure of working capital management.

In Table 8B, we perform another robustness test. Here, we use age group dummy variables for three age groups of the executives. So, we use two dummies each in three models. In Model 1, the coefficient of the *Executive Age* dummy for executive age < 50 years is negative and highly significant, indicating that compared to the executives older than 60 years (the reference group), younger executives exhibit shorter cash conversion cycles. Similarly, the coefficient of the *Executive Age* dummy for executive age between 50 and 60 years is also negative and significant but the coefficient for this age group is smaller than the coefficient for the youngest age group, suggesting that the cash conversion cycle increases with executive age. In Model 2, our reference group for dummy variables is the executive age less than 50 years. The positive and significant coefficients for dummy variables in the other two groups in which the coefficient for oldest age group is bigger than the age group of executives between 50 and 60 years are consistent with the notion that the cash conversion cycle increases with executive age. Similarly, in Model 3, the reference dummy is the age group of the executives between 50 and 60 years. In this model, the coefficient for the dummy variable for executives less than 50 years is negative and significant whereas the coefficient for the dummy variable for executives

Table 8A

Robustness test: alternate dependent variable

Alternative dependent variable (all executives)		
	(Current assets – Current liabilities)/Total assets	(Current assets – Accounts payable)/Total assets
	Model 1	Model 2
	Standard errors are clustered at firm and executive level	Standard errors are clustered at firm and executive level
<i>Executive Age</i>	0.0011*** (0.000)	0.0007*** (0.0002)
<i>Sales Growth (t–1)</i>	–0.0045*** (0.001)	–0.0029* (0.0015)
<i>Sales Volatility</i>	–0.0000*** (0.000)	0.0000 (0.0000)
<i>Gross Profit Margin (t–1)</i>	–0.1054*** (0.010)	0.0080 (0.0151)
<i>Operating Cash Flow (t–1)</i>	0.1889*** (0.013)	0.0256 (0.0197)
<i>Firm Size (t–1)</i>	–0.0127*** (0.001)	–0.0406*** (0.0017)
<i>Market-to-Book (t–1)</i>	–0.0096*** (0.002)	0.0248*** (0.0020)
<i>Leverage (t–1)</i>	–0.1266*** (0.012)	–0.2421*** (0.0178)
<i>Tangibility (t–1)</i>	–0.1394*** (0.010)	–0.2929*** (0.0156)
<i>Distress Dummy</i>	0.0206*** (0.003)	0.0618*** (0.0045)
Constant	0.2531*** (0.025)	0.7790*** (0.0500)
R^2	0.401	0.5891
Adj R^2	0.4008	0.5889
Observations	25,439	25,439
Industry controls	Yes	Yes
Year controls	Yes	Yes

This table presents the multivariate regressions for robustness. The dependent variable is *Net Working Capital Ratio* (defined in two different ways) and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Model 1 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

Table 8B

Robustness test: executive age as a dummy variable

Cash conversion cycle			
All executives			
	Model 1	Model 2	Model 3
	Standard errors are clustered at firm level	Standard errors are clustered at firm level	Standard errors are clustered at firm level
Dummy (<i>Executive Age</i> < 50)	−13.8695*** (−4.6330)		−7.6840*** (−2.8296)
Dummy (<i>Executive Age</i> between 50 and 60)	−6.1855*** (−2.6773)	7.6840*** (2.8296)	
Dummy (<i>Executive Age</i> > 60)		13.8695*** (4.6330)	6.1855*** (2.6773)
<i>Sales Growth</i> (<i>t</i> −1)	−5.5476*** (−4.2016)	−5.5476*** (−4.2016)	−5.5476*** (−4.2016)
<i>Sales Volatility</i>	−0.0011 (−0.7621)	−0.0011 (−0.7621)	−0.0011 (−0.7621)
<i>Gross Profit Margin</i> (<i>t</i> −1)	−221.1728*** (−13.5034)	−221.1728*** (−13.5034)	−221.1728*** (−13.5034)
<i>Operating Cash Flow</i> (<i>t</i> −1)	135.5082*** (8.4199)	135.5082*** (8.4199)	135.5082*** (8.4199)
<i>Firm Size</i> (<i>t</i> −1)	−1.4256 (−0.9536)	−1.4256 (−0.9536)	−1.4256 (−0.9536)
<i>Market-to-Book</i> (<i>t</i> −1)	−5.7131*** (−3.5294)	−5.7131*** (−3.5294)	−5.7131*** (−3.5294)
<i>Leverage</i> (<i>t</i> −1)	7.7373 (0.6005)	7.7373 (0.6005)	7.7373 (0.6005)
<i>Tangibility</i> (<i>t</i> −1)	−69.3450*** (−6.1301)	−69.3450*** (−6.1301)	−69.3450*** (−6.1301)
<i>Distress Dummy</i>	1.8250 (0.5637)	1.8250 (0.5637)	1.8250 (0.5637)
Constant	166.7270*** (4.9846)	152.8575*** (4.5663)	160.5415*** (4.8028)
R^2	0.3328	0.3328	0.3328
Adj R^2	0.3310	0.3310	0.3310
Observations	25,439	25,439	25,439
Industry controls	Yes	Yes	Yes
Year controls	Yes	Yes	Yes

This table presents the multivariate regressions for robustness using three executive age groups as dummy variables. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Model 1 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

Table 8C

Robustness test: comparing young executive and old executive groups

Executives groups		
	Model 1	Model 2
Young CEO and Young CFO (Both < 50 years) (Dummy = 1 if both CEO & CFO are young)	-10.8393** (4.9702)	
Old CEO and Old CFO (Both > 60 years) (Dummy = 1 if both CEO & CFO are old)		20.8983** (8.8354)
<i>Sales Growth</i> ($t-1$)	-2.6125 (4.8320)	-2.8748 (4.8282)
<i>Sales Volatility</i>	-0.0030*** (0.0010)	-0.0030*** (0.0010)
<i>Gross Profit Margin</i> ($t-1$)	-251.2096*** (14.3740)	-251.7537*** (14.3336)
<i>Operating Cash Flow</i> ($t-1$)	108.5879*** (19.0060)	108.7052*** (18.9635)
<i>Firm Size</i> ($t-1$)	-1.1863 (1.2314)	-0.9895 (1.2289)
<i>Market-to-Book</i> ($t-1$)	-3.3853* (1.9551)	-3.6058* (1.9505)
<i>Leverage</i> ($t-1$)	9.9985 (12.8640)	11.0766 (12.8468)
<i>Tangibility</i> ($t-1$)	-45.7226*** (10.3123)	-44.8673*** (10.3578)
Distress Dummy	2.4335 (3.5562)	2.1324 (3.5631)
Constant	102.9722*** (17.2112)	101.3011*** (17.1966)
R^2	0.3693	0.3694
Adj R^2	0.3682	0.3683
Observations	5,506	5,506
Industry controls	Yes	Yes
Year controls	Yes	Yes

This table presents the multivariate regressions for robustness using three executive age groups as dummy variables. The dependent variable is *Cash Conversion Cycle* and we include standard determinants along with *Executive Age*. The accounting variables are from Compustat and executive specific variables are obtained from the ExecuComp files on Compustat. The full sample consists of observations for both CEOs and CFOs. The sample consists of annual observations between 1993 and 2014. ***, ** and * represent significance at the 0.01, 0.05 and 0.10 levels, respectively. The values in parentheses for Model 1 are heteroskedasticity robust standard errors. Variable definitions are presented in the Appendix.

greater than 60 years is significant and positive. These results provide further evidence on the inverse relationship between cash conversion cycle and executive age.

Additionally, we create dummy variables for both young executives (both CEO and CFO younger than 50 years in a given firm year) and for both old executives (both CEO and CFO older than 60 years in a given firm year) and examine such combined age effect on cash conversion cycle. We report these results in Table 8C. We find that the younger group of executives is significantly negatively related with the cash conversion cycle, whereas the older group of executives is significantly positively related with cash conversion cycle, strongly corroborating our finding that older executives are risk averse and invest more heavily in working capital compared to their younger counterparts.

Putting it all together, our findings are consistent with the notion that older executives are more risk averse as compared to their younger counterparts and this is reflected in their short-term financial management policies. Furthermore, these conservative policies may reduce firm value as several existing studies have pointed towards the detrimental impact of excessive investment in working capital.

Furthermore, we also perform regressions with industry and time controls, quantile regression and Fama–Macbeth regressions to further corroborate our results in Table 2. We add industry and year dummies in our regression model to rule out the impact of potential unobservable time-invariant, industry-specific and year-specific effects on the measures of working capital management. We also perform quantile regressions which have the advantage that their estimates are more robust against outliers in the dependent variable. Finally, we analyse our hypothesis with Fama–MacBeth regressions to correct for any potential cross-sectional dependence in residuals, which biases the results in favour of the acceptance of the hypothesis if left untreated. Thus, our research design addresses potential endogeneity concerns. When we execute regressions with industry and time controls, quantile regressions and Fama–Macbeth regressions for CEOs only and CFOs only samples with controls as in Table 2, we find qualitatively similar results as described above for Table 2. We do not report these results for brevity. These findings confirm the notion that firms led by older CEOs and older CFOs exhibit the influence of risk aversion in their corporate policies.

5. Conclusion

Some of the prior studies have established that the older executives pursue risk-averse strategies. However, these studies do not focus on short-term policies like working capital management. We add to the literature by connecting short-term financial policies to executive age. We are the first to focus on the effect of executive age on working capital management. Our finding that older executives invest more heavily in working capital is consistent with the view that older executives are more risk averse. Our results have been established using sophisticated econometric techniques and with heteroskedasticity-adjusted standard errors clustered at the firm-level.

An interesting implication of our study is that it provides new evidence and reasoning in identifying a subset of executives (older executives) who invest more heavily in working capital which may make their firm less profitable. Our findings not only initiate new avenues in the research relating short-term financial decision making to executive characteristics but also assist firms in choosing the right type of executives for their firms.

References

- Abdulla, Y., V. A. Dang, and A. Khurshed, 2015, The use of trade credit by public and private firms: an empirical investigation, Available at SSRN 2534029.
- Adhikari, B., 2012, *Gender differences in corporate financial decisions and performance*, Working Paper (University of Alabama).
- Adhikari, H. P., S. B. Bulmash, M. W. Krolkowski, and N. B. Sah, 2015, Dynamics of CEO compensation: old is gold, *The Quarterly Review of Economics and Finance* 57, 191–206.
- Agarwal, R., and M. Gort, 1996, The evolution of markets and entry, exit and survival of firms, *The Review of Economics and Statistics* 78, 489–498.
- Agarwal, R., and M. Gort, 2002, Firm and product life cycles and firm survival, *The American Economic Review* 92, 184–190.
- Aktas, N., E. Croci, and D. Petmezas, 2015, Is working capital management value-enhancing? Evidence from firm performance and investments, *Journal of Corporate Finance* 30, 98–113.
- Albert, S. M., and J. Duffy, 2012, Differences in risk aversion between young and older adults, *Neuroscience and Neuroeconomics* 2012, <https://doi.org/10.2147/NAN.S27184>
- Ali, A., and W. Zhang, 2015, CEO tenure and earnings management, *Journal of Accounting and Economics* 59, 60–79.
- Andreou, P. C., C. Louca, and A. P. Petrou, 2017, CEO age and stock price crash risk, *Review of Finance* 21, 1287–1325.
- Arora, M., and S. Kumari, 2015, Risk taking in financial decisions as a function of age, gender: mediating role of loss aversion and regret, *International Journal of Applied Psychology* 5, 83–89.
- Arrow, K., 1962, Economic welfare and the allocation of resources for invention, in: *The Rate and Direction of Inventive Activity: Economic and Social Factors* (Princeton University Press, Princeton, NJ), 609–626.
- Baños-Caballero, S., P. J. García-Teruel, and P. Martínez-Solano, 2010, Working capital management in SMEs, *Accounting and Finance* 50, 511–527.
- Baños-Caballero, S., P. J. García-Teruel, and P. Martínez-Solano, 2014, Working capital management, corporate performance, and financial constraints, *Journal of Business Research* 67, 332–338.
- Barker, V. L. III, and G. C. Mueller, 2002, CEO characteristics and firm R&D spending, *Management Science* 48, 782–801.
- Bertrand, M., and S. Mullainathan, 2003, Enjoying the quiet life? Corporate governance and managerial preferences, *Journal of Political Economy* 111, 1043–1075.
- Bertrand, M., and A. Schoar, 2003, Managing with style: the effect of managers on firm policies, *Quarterly Journal of Economics* 118, 1169–1208.
- Bhagat, S., and I. Welch, 1995, Corporate research & development investments international comparisons, *Journal of Accounting and Economics* 19, 443–470.
- Bhattacharya, H., 2001, *Working Capital Management: Strategies and Techniques* (Prentice Hall, New Delhi).

- Billett, M. T., T. H. D. King, and D. C. Mauer, 2007, Growth opportunities and the choice of leverage, debt maturity, and covenants, *The Journal of Finance* 62, 697–730.
- Cain, M. D., and S. B. McKeon, 2016, CEO personal risk-taking and corporate policies, *Journal of Financial and Quantitative Analysis* 51, 139–164.
- Cameron, A. C., and D. L. Miller, 2015, A practitioner's guide to cluster-robust inference, *Journal of Human Resources* 50, 317–372.
- Cassell, C. A., S. X. Huang, J. M. Sanchez, and M. D. Stuart, 2012, Seeking safety: the relation between CEO inside debt holdings and the riskiness of firm investment and financial policies, *Journal of Financial Economics* 103, 588–610.
- Chakraborty, A., S. Sheikh, and N. Subramanian, 2007, Termination risk and managerial risk taking, *Journal of Corporate Finance* 13, 170–188.
- Chatterjee, A., and D. C. Hambrick, 2007, It's all about me: narcissistic chief executive officers and their effects on company strategy and performance, *Administrative Science Quarterly* 52, 351–386.
- Chava, S., and A. Purnanandam, 2010, CEOs versus CFOs: incentives and corporate policies, *Journal of Financial Economics* 97, 263–278.
- Child, J., 1974, Managerial and organizational factors associated with company performance, *Journal of Management Studies* 11, 175–189.
- Coles, J. L., N. D. Daniel, and L. Naveen, 2006, Managerial incentives and risk-taking, *Journal of Financial Economics* 79, 431–468.
- Davidson, W. N. III, B. Xie, W. Xu, and Y. Ning, 2007, The influence of executive age, career horizon and incentives on pre-turnover earnings management, *Journal of Management and Governance* 11, 45–60.
- Deloof, M., 2003, Does working capital management affect profitability of Belgian firms?, *Journal of Business Finance and Accounting* 30, 573–587.
- Deloof, M., and M. Jegers, 1996, Trade credit, product quality, and intragroup trade: some European evidence, *Financial Management* 25, 33–43.
- Diamond, D. W., 1989, Reputation acquisition in debt markets, *Journal of Political Economy* 97, 828–862.
- Ding, S., A. Guariglia, and J. Knight, 2013, Investment and financing constraints in China: does working capital management make a difference?, *Journal of Banking and Finance* 37, 1490–1507.
- Dittmar, A. K., and R. Duchin, 2010, The dynamics of cash, Working Paper (Ross Business School).
- Dohmen, T., A. Falk, B. H. Golsteyn, D. Huffman, and U. Sunde, 2017, Risk attitudes across the life course, *The Economic Journal* 127, F95–F116.
- Dunn, P., and L. Cheatham, 1993, Fundamentals of small business financial management for start up, survival, growth, and changing economic circumstances, *Managerial Finance* 19, 1–13.
- Eberhart, A. C., W. F. Maxwell, and A. R. Siddique, 2004, An examination of long-term abnormal stock returns and operating performance following R&D increases, *The Journal of Finance* 59, 623–650.
- Ericson, R., and A. Pakes, 1995, Markov-perfect industry dynamics: a framework for empirical work, *The Review of Economic Studies* 62, 53–82.
- Faccio, M., M.-T. Marchica, and R. Mura, 2016, CEO gender, corporate risk-taking, and the efficiency of capital allocation, *Journal of Corporate Finance* 39, 193–209.
- Faulkender, M., and R. Wang, 2006, Corporate financial policy and the value of cash, *The Journal of Finance* 61, 1957–1990.
- Fazzari, S., and B. Petersen, 1993, Working capital and fixed investment: new evidence on financing constraints, *RAND Journal of Economics* 24, 328–342.

- Francis, B., I. Hasan, J. C. Park, and Q. Wu, 2015, Gender differences in financial reporting decision making: evidence from accounting conservatism, *Contemporary Accounting Research* 32, 1285–1318.
- García-Teruel, P. J., and P. Martínez-Solano, 2007, Effects of working capital management on SME profitability, *International Journal of Managerial Finance* 3, 164–177.
- Gibbons, R., and K. J. Murphy, 1992, Does executive compensation affect investment?, NBER Working Paper No. 4135.
- Gill, A., N. Biger, and N. Mathur, 2010, The relationship between working capital management and profitability: evidence from the United States, *Business and Economics Journal* 10, 1–9.
- Grady, C. L., A. R. McIntosh, B. Horwitz, and S. I. Rapoport, 2000, Age-related changes in the neural correlates of degraded and non-degraded face processing, *Cognitive Neuropsychology* 217, 165–186.
- Hambrick, D. C., 2007, Upper echelons theory: an update, *Academy of Management Review* 32, 334–343.
- Hambrick, D. C., and P. A. Mason, 1984, Upper echelons: the organization as a reflection of its top managers, *Academy of Management Review* 9, 193–206.
- Harman, D., 1991, The aging process: major risk factor for disease and death, *Proceedings of the National Academy of Sciences* 88, 5360–5363.
- Hayajneh, O. S., and F. L. A. Yassine, 2011, The impact of working capital efficiency on profitability: an empirical analysis on Jordanian manufacturing firms, *International Research Journal of Finance and Economics* 66, 67–69.
- Hill, M. D., G. W. Kelly, and M. J. Highfield, 2010, Net operating working capital behavior: a first look, *Financial Management* 39, 783–805.
- Hirshleifer, D., and A. V. Thakor, 1992, Managerial conservatism, project choice, and debt, *The Review of Financial Studies* 5, 437–470.
- Holmström, B., 1999, The firm as a subeconomy, *Journal of Law, Economics, and Organization* 15, 74–102.
- Huang, J., and D. J. Kigen, 2013, Gender and corporate finance: are male executives overconfident relative to female executives?, *Journal of Financial Economics* 108, 822–839.
- Huang, H. W., E. Rose-Green, and C. C. Lee, 2012, CEO age and financial reporting quality, *Accounting Horizons* 26, 725–740.
- Hunter, K., and S. Kemp, 2004, The personality of e-commerce investors, *Journal of Economic Psychology* 25, 529–537.
- Hunter, D. M., S. A. Mansi, and N. B. Sah, 2015, Cash is queen: female CEOs' propensity to hoard cash, Working Paper (University of South Florida).
- Huynh, P. D., and J. Su, 2010, The relationship between working capital management and profitability: a Vietnamese case, *International Research Journal of Finance and Economics* 64, 59–67.
- Jensen, M. C., and W. H. Meckling, 1976, Theory of the firm: managerial behavior, agency costs and ownership structure, *Journal of Financial Economics* 3, 305–360.
- Jovanovic, B., 1982, Selection and the evolution of industry, *Econometrica* 50, 649–670.
- Kahneman, D., and A. Tversky, 1979, Prospect theory: an analysis of decision under risk, *Econometrica* 47, 263–291.
- Kieschnick, R., M. Laplante, and R. Moussawi, 2013, Working capital management and shareholders' wealth, *Review of Finance* 17, 1827–1852.
- Kim, Y. H., and K. H. Chung, 1990, An integrated evaluation of investment in inventory and credit: a cash flow approach, *Journal of Business Finance and Accounting* 17, 381–389.

- Kini, O., and R. Williams, 2012, Tournament incentives, firm risk, and corporate policies, *Journal of Financial Economics* 103, 350–376.
- Kothari, S. P., T. E. Laguerre, and A. J. Leone, 2002, Capitalization versus expensing: evidence on the uncertainty of future earnings from capital expenditures versus R&D outlays, *Review of Accounting Studies* 7, 355–382.
- Kuhnen, C. M., and B. Knutson, 2011, The influence of affect on beliefs, preferences, and financial decisions, *Journal of Financial and Quantitative Analysis* 46, 605–626.
- Li, X., A. Low, and A. K. Makhija, 2017, Career concerns and the busy life of the young CEO, *Journal of Corporate Finance* 47, 88–109.
- Loughran, T., and J. R. Ritter, 2004, Why has IPO underpricing changed over time?, *Financial Management* 33, 5–37.
- Love, I., L. A. Preve, and V. Sarria-Allende, 2007, Trade credit and bank credit: evidence from recent financial crises, *Journal of Financial Economics* 83, 453–469.
- MacCrimmon, K. R., and D. A. Wehrung, 1986, *Taking Risks: The Management of Uncertainty* (Free Press, New York, NY).
- Mather, M., N. Mazar, M. A. Gorlick, N. R. Lighthall, J. Burgeno, A. Schoeke, and D. Ariely, 2012, Risk preferences and aging: the ‘certainty effect’ in older adults’ decision making, *Psychology and Aging* 27, 801–816.
- Morin, R. A., and A. F. Suarez, 1983, Risk aversion revisited, *The Journal of Finance* 38, 1201–1216.
- Myers, S. C., and N. S. Majluf, 1984, Corporate financing and investment decisions when firms have information that investors do not have, *Journal of Financial Economics* 13, 187–221.
- Nobanee, H., M. Abdullatif, and M. AlHajjar, 2011, Cash conversion cycle and firm’s performance of Japanese firms, *Asian Review of Accounting* 19, 147–156.
- Olsen, B. C., S. R. Sisodiya, and J. Swisher, 2016, A note on assessing the relation between CEO characteristics and stock performance: alpha above replacement, *Accounting and Finance* 56, 787–802.
- Padachi, K., 2006, Trends in working capital management and its impact on firms’ performance: an analysis of Mauritian small manufacturing firms, *International Review of Business Research Papers* 2, 45–58.
- Petersen, M. A., 2009, Estimating standard errors in finance panel data sets: comparing approaches, *The Review of Financial Studies* 22, 435–480.
- Petersen, M. A., and R. G. Rajan, 1997, Trade credit: theories and evidence, *The Review of Financial Studies* 10, 661–691.
- Prendergast, C., and L. Stole, 1996, Impetuous youngsters and jaded old-timers: acquiring a reputation for learning, *Journal of Political Economy* 104, 1105–1134.
- Roberts, S., and I. Rosenberg, 2006, Nutrition and aging: changes in the regulation of energy metabolism with aging, *Physiological Reviews* 86, 651–667.
- Scharfstein, D. S., and J. C. Stein, 1990, Herd behavior and investment, *The American Economic Review* 80, 465–479.
- Schiff, M., and Z. Lieber, 1974, A model for the integration of credit and inventory management, *The Journal of Finance* 29, 133–140.
- Serfling, M. A., 2014, CEO age and the riskiness of corporate policies, *Journal of Corporate Finance* 25, 251–273.
- Shin, H. H., and L. Soenen, 1998, Efficiency of working capital management and corporate profitability, *Financial Practice and Education* 8, 37–45.
- Stiglitz, J. E., and A. Weiss, 1981, Credit rationing in markets with imperfect information, *The American Economic Review* 71, 393–410.
- Sundaram, R. K., and D. L. Yermack, 2007, Pay me later: inside debt and its role in managerial compensation, *The Journal of Finance* 62, 1551–1588.

- Vroom, V. H., and B. Pahl, 1971, Relationship between age and risk taking among managers, *Journal of Applied Psychology* 55, 399–405.
- Wang, Y. J., 2002, Liquidity management, operating performance, and corporate value: evidence from Japan and Taiwan, *Journal of Multinational Financial Management* 12, 159–169.
- Yim, S., 2013, The acquisitiveness of youth: CEO age and behavior, *Journal of Financial Economics* 108, 250–273.
- Zwiebel, J., 1995, Corporate conservatism and relative compensation, *Journal of Political Economy* 103, 1–25.

Appendix

Cash Conversion Cycle: The time lag between the collection of revenue from the sales of finished products or services and the expenditure for the purchase of raw materials. A longer *Cash Conversion Cycle* increases the investment in working capital. This is our dependent variable in the majority of our specifications.

Days Inventory Outstanding: Defined as $[\text{inventories} \times 365] / \text{cost of sales}$.

Longer *Days Inventory Outstanding* increases investment in working capital.

Days Receivables Outstanding: Defined as $[\text{Receivables} \times 365] / \text{Sales}$. Longer *Days Receivables Outstanding* increases investment in working capital.

Days Payables Outstanding: Defined as $[\text{Accounts Payables} \times 365] / \text{Purchases}$.

Longer *Days Payables Outstanding* decreases investment in working capital.

Executive Age: Age of the current CEO or CFO of the firm depending on the executive considered in that firm-year observation.

CEO age: Age of the current CEO of the firm.

CFO age: Age of the current CFO of the firm.

Sales growth: Defined as $((\text{This year's sales} - \text{Previous year's sales}) / \text{Previous year's sales})$. Existing literature (e.g., Petersen and Rajan, 1997) contends that sales growth affects the working capital requirements of a firm, but the direction of the relationship is not clear. We include lagged sales growth as a control variable following Love *et al.* (2007).

Sales Volatility: Defined as the standard deviation of a firm's annual net sales over a rolling five-year period prior to each of the sample years. We include this variable as control as the companies with more volatile revenues will have more difficulty predicting their short-term financing needs and may have to depend on accounts payable to maintain their cash flows (Hill *et al.*, 2010).

Gross Profit Margin: Defined as $(\text{Sales} - \text{Cost of goods sold}) / \text{Sales}$. We include this variable as control because we anticipate that the profit margin directly influences working capital needs. As in the case of most other variables, we use lagged gross profit margin as a control variable in our regressions.

Operating Cash Flows: A positive operating cash flow allows companies to follow more conservative working capital policy, whereas a negative

operating cash flow makes companies rely on other sources, so we include this variable as our control. Following existing literature (e.g., Andreou *et al.*, 2017), we use lagged operating cash flows to mitigate endogeneity concerns.

Firm Size: The natural logarithm of market value of equity in inflation-adjusted 2002 dollars. We include firm size as it captures the accessibility of a firm to the capital market.

Market-to-Book: The ratio of the market value of equity to the book value of equity. Market value of a firm is measured at the beginning of the fiscal year. It captures the degree of asymmetric information. We use lagged market-to-book ratio as a control variable in our regressions.

Leverage: The ratio of debt to total assets. Leverage is a commonly used control variable in finance and accounting literature and hence we use lagged leverage ratio as a control variable in our regressions.

Tangibility: Tangibility is defined as the tangible fixed assets scaled by total assets. This variable will help control for the illiquid assets of a firm. Hence, we use lagged tangibility as a control variable in our regressions.

Distress dummy: As distressed firms can have only limited levels of trade credit compared to non-distressed firms, distressed firms are expected to invest less in operating working capital; instead, they depend on collecting receivables or on stretching the credit terms provided by suppliers. We include financial distress dummy in our model specifications. If the firm is distressed (decided using the Kaplan–Zingales Index score) then the Distress dummy is one, otherwise zero.

Firm Age: The natural logarithm of the age of the firm. We include *Firm Age* as a control as it captures the strength of the firm's internal control (Huang *et al.*, 2012) and hence can influence short-term financial management.

Executive Gender: Dummy variable that is equal to one if the executive is a female and zero otherwise.

Executive Tenure: Number of years the executive has spent at the firm in that particular post.

Net Working Capital Ratio: Difference between current assets and current liabilities scaled by total assets. We also use alternative definition of *Net Working Capital Ratio* as (current assets – accounts payable)/total assets.